OptiCrop: Smart Agricultural Production Optimization Engine

Coding & Solution

Coding & Solution

This project implements an AI-powered smart farming solution that recommends suitable crops based on soil and environmental conditions. The application uses Machine Learning for crop prediction and provides fertilizer recommendations, weather information, and a user-friendly web interface. The code is modular, well-structured, reusable, and follows standard coding practices to ensure maintainability and scalability.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub / GitLab or hosted link here]
Programming Language(s)	[e.g., Python, HTML, CSS, JavaScript]
Framework(s) Used	[e.g.,Flask,Scikit-learn,Pandas,Numpy]
Key Features Implemented	Crop Recommendation using Machine Learning ,Soil Parameter Analysis ,Telugu Crop Name Display ,Crop Image Display ,User-Friendly Web Interface ,Prediction Result Visualization
Pending / Incomplete Features	Weather API Integration ,Fertilizer Recommendation System ,Farmer History Tracking ,Mobile Application Support
Setup / Run Instructions	Install Python dependencies from requirements.txt ,Place Crop_recommendation.csv in project directory ,Load model.pkl ,Run app.py ,Open browser and access Flask application

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (if uploaded to GitHub)

Additional Notes / Comments: